

Impact of Accounts Receivable & Payable Days on Working Capital Efficiency

Domain: Corporate Finance (FMCG Sector) | Tools: EViews, SPSS | Data: 226 Indian FMCG Companies, 2011–2024 (Panel Data, CMIE ProwessIQ)

Summary

Objective

Examine how Accounts Payable Days (AP_DAYS) and Accounts Receivable Days (AR_DAYS) impact the Cash Conversion Cycle (CCC) of Indian FMCG companies, to identify which lever — collections or payables — offers more control over working capital efficiency.

Concepts / Methods Used

Panel Data Econometrics: Pooled OLS (POLS), Breusch-Pagan Lagrange Multiplier Test, Random Effects Model (REM), Hausman Specification Test, Fixed Effects Model (FE). Diagnostics: Correlation matrix, VIF (multicollinearity check), Z-score normalization, Kolmogorov–Smirnov normality test.

Key Output

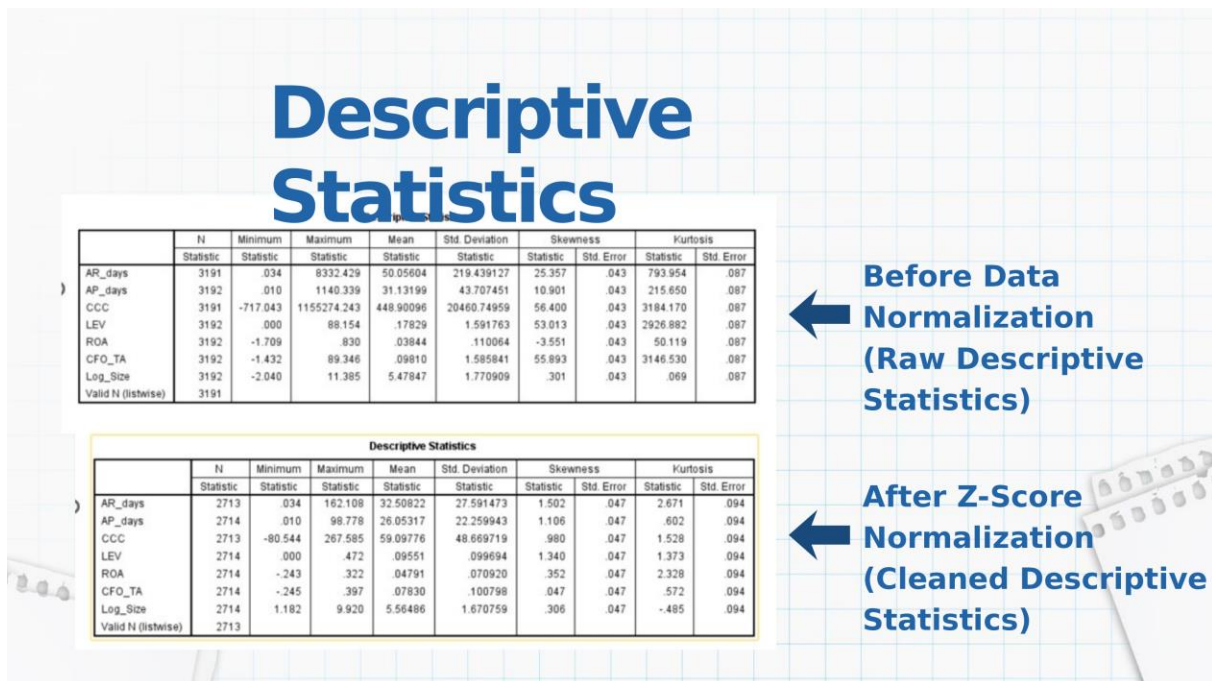
- Final dataset: 226 Indian FMCG companies, unbalanced panel, 2,713 firm-year observations (2011–2024) after data cleaning
- AR_DAYS model (Random Effects, selected via Hausman $p=0.5081$): $CCC = 2.50 + 1.02 \cdot AR_DAYS - 17.14 \cdot CFO_TA + 4.46 \cdot \log(TA) - 28.69 \cdot ROA + 8.38 \cdot LEV$ ($R^2 = 0.383$)
- AP_DAYS model (Fixed Effects, selected via Hausman $p=0.0038$): $CCC = 29.22 - 0.75 \cdot AP_DAYS - 26.24 \cdot CFO_TA + 9.78 \cdot \log(TA) - 81.32 \cdot ROA + 8.96 \cdot LEV$ ($R^2 = 0.734$)
- AR_Days has a near 1:1 positive relationship with CCC ($\beta=1.02$, $p<0.001$) — receivables are the dominant driver of cash conversion cycle length
- AP_Days has a significant negative relationship with CCC ($\beta=-0.75$, $p<0.001$) — delaying payables shortens the cycle, but with less than 1:1 leverage
- All VIF values ≈ 1 — no multicollinearity concern; all variables retained

Business Actionables

- Prioritize receivables collection efficiency first — AR_Days has the strongest, most direct effect on cash tied up in the business
- Use payables extension as a secondary, not primary, lever — it helps but moves the cycle less per day than receivables do
- Firm size and profitability (ROA) also significantly affect CCC — working capital targets should be benchmarked within similar size/profitability peer groups, not company-wide averages
- Because AP_Days requires a Fixed Effects model (firm-specific factors matter) while AR_Days does not, payables policy should be tailored per firm/sub-sector rather than applied as a uniform rule

Detailed Analysis — Actual EViews/SPSS Output, Explained, With Actionables

Output 1 — Descriptive Statistics (Raw and Z-Score Normalized)



Actual SPSS output: descriptive statistics before and after Z-score normalization

What this is: Summary statistics (mean, min, max, standard deviation, skewness, kurtosis) for CCC, AP_Days, AR_Days, ROA, LEV, CFO_TA, and firm size, computed on ~2,713–2,714 valid firm-year observations, shown both before and after Z-score normalization and outlier trimming.

Layman meaning: The average FMCG firm in this sample takes about 59 days to convert its investments back into cash (mean CCC $\approx$ 59), but with very high variability (std. dev. $\approx$ 48.7 days, and some firms show negative CCC or CCC over 250 days). Average AR_Days ($\approx$ 32.5) is higher than average AP_Days ($\approx$ 26), meaning firms typically wait longer to collect from customers than they take to pay suppliers — a working capital drag by default.

Actionable for the user: The wide spread in CCC signals that a single company-wide working capital target would be unrealistic — targets should be set per size/profitability band, which is echoed directly in the paper's practical implications.

Output 2 — VIF (Multicollinearity Check)

VIF

All VIF values are low (around 1), indicating no multicollinearity; therefore, all variables were retained in the model.

Coefficients ^a										
Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations			Collinearity Statistics	
	B	Std. Error	Beta			Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)	33.603	2.682		.000					
	AR_Days	1.149	.027	.651	.000	.600	.639	.609	.874	1.144
	AP_Days	-.697	.032	-.318	.000	-.173	-.388	-.308	.938	1.066
	LEV	-.065	7.071	.000	.993	.044	.000	.000	.945	1.059
	ROA	-17.884	11.337	-.026	.115	-.111	-.030	-.022	.726	1.377
	CFO_TA	-36.823	7.564	-.076	.000	-.040	-.093	-.069	.808	1.238
	Log_Total Assets	1.802	.437	.062	.000	.186	.079	.058	.881	1.135

a. Dependent Variable: CCC

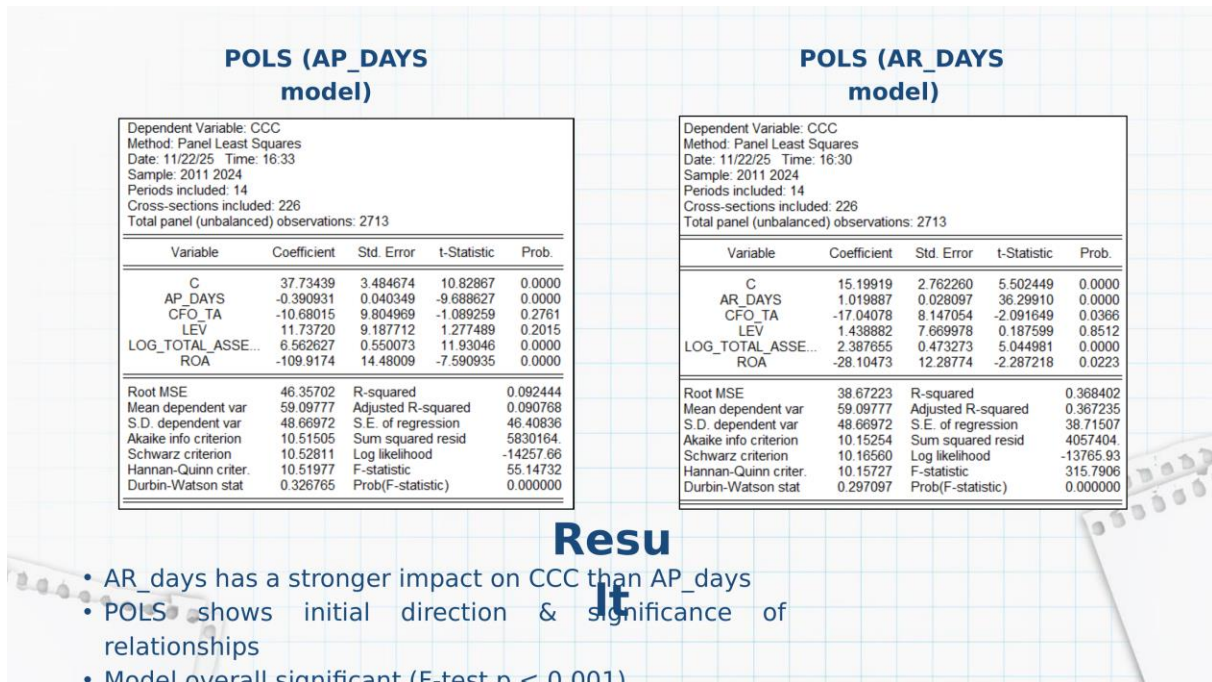
Actual SPSS output: Collinearity Statistics — Tolerance and VIF

What this is: Variance Inflation Factor test run across all control variables (ROA, LEV, CFO_TA, firm size) alongside AP_Days/AR_Days to check whether any predictors are redundant with each other.

Layman meaning: All VIF values came out close to 1, far below any concerning threshold (5 or 10). None of the variables are secretly measuring the same thing — every variable in the model adds distinct information.

Actionable for the user: This is a validity checkpoint: it confirms the regression coefficients reported later (for AP_Days and AR_Days) are trustworthy on their own terms, not distorted by overlapping predictors.

Output 3 — Pooled OLS (POLS): Initial Models for AP_Days and AR_Days



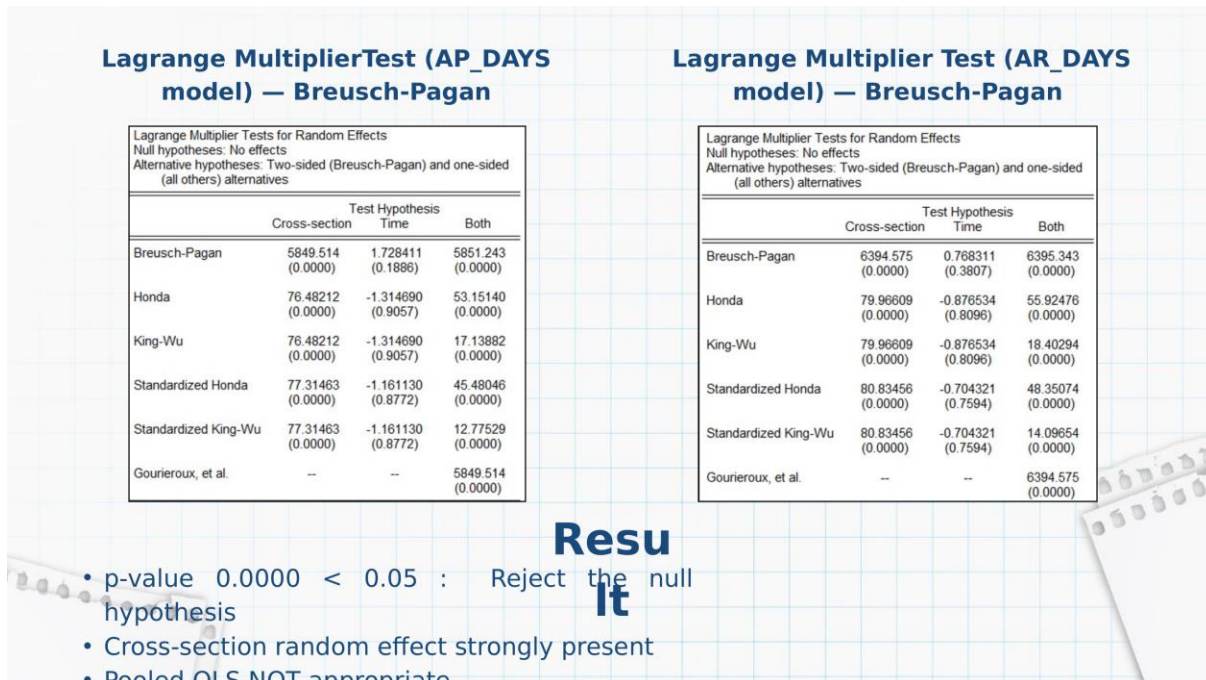
Actual EViews output: Panel Least Squares regression for both AP_Days and AR_Days models

What this is: The starting-point regression, treating all 2,713 firm-year observations as one pooled sample (ignoring that observations are grouped by firm). Two separate models were run — one isolating AP_Days, one isolating AR_Days — against CCC.

Layman meaning: Even at this first, simplest pass, both relationships are already highly significant ($p < 0.001$): AP_Days has a negative coefficient (-0.39) and AR_Days has a positive coefficient (1.02) on CCC. But POLS treats every company-year as independent, which is usually not a safe assumption for company financial data — hence the need for the next diagnostic step.

Actionable for the user: POLS gives a first directional read, but a business user should not stop here — firm-level differences (culture, sector, management) need to be tested for before trusting the number, which is exactly what the next two tests do.

Output 4 — Breusch-Pagan Lagrange Multiplier Test



Actual EViews output: Lagrange Multiplier Test for Random Effects (both models)

What this is: A statistical test of whether firm-specific effects exist in the data (i.e., whether some companies are systematically different from others in ways that a pooled model would ignore).

Layman meaning: The test result is decisively significant ($p < 0.0001$) for both models. In plain terms: yes, individual companies do have their own persistent characteristics affecting their cash conversion cycle — a simple pooled regression is the wrong tool, and a model that accounts for company-level differences is required.

Actionable for the user: This justifies why the analysis moved to a Random Effects Model next — it's not a technical formality, it's evidence-based: skipping this step would have meant reporting AP_Days/AR_Days effects that ignore real company-to-company differences and could mislead.

Output 5 — Random Effects Model (REM)

Random Effect Model (REM) Results — AP Days & AR Days				
REM Output for AP Days (Independent Variable)				
Dependent Variable: CCC Method: Panel EGLS (Cross-section random effects) Date: 11/19/25 Time: 19:41 Sample: 2011 2024 Periods included: 14 Cross-sections included: 226 Total panel (unbalanced) observations: 2713 Swamy and Arora estimator of component variances				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	37.78803	5.953882	6.346787	0.0000
AP_DAYS	-0.687198	0.043390	-15.83783	0.0000
CFO_TA	-26.48130	6.107736	-4.335697	0.0000
LOG_TOTAL_ASSE...	8.357205	0.923409	9.050384	0.0000
ROA	-84.34089	10.36388	-8.137966	0.0000
LEV	8.184383	7.506034	1.090374	0.2756
Effects Specification				
		S.D.	Rho	
Cross-section random		38.33115	0.6807	
Idiosyncratic random		26.25168	0.3193	
Weighted Statistics				
Root MSE	26.34916	R-squared	0.136375	
Mean dependent var	11.37429	Adjusted R-squared	0.134780	
S.D. dependent var	28.45462	S.E. of regression	26.37834	
Sum squared resid	1883576	F-statistic	85.49258	
Durbin-Watson stat	0.979409	Prob(F-statistic)	0.000000	
Unweighted Statistics				
R-squared	0.066587	Mean dependent var	59.09777	
Sum squared resid	5996271	Durbin-Watson stat	0.307656	
REM Output for AR Days (Independent Variable)				
Dependent Variable: CCC Method: Panel EGLS (Cross-section random effects) Date: 11/19/25 Time: 19:40 Sample: 2011 2024 Periods included: 14 Cross-sections included: 226 Total panel (unbalanced) observations: 2713 Swamy and Arora estimator of component variances				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.500236	4.960892	0.503989	0.6143
AR_DAYS	1.022988	0.026958	37.94680	0.0000
CFO_TA	-17.13555	5.143813	-3.331294	0.0009
LOG_TOTAL_ASSE...	4.463409	0.783574	5.696221	0.0000
ROA	-28.69305	8.914381	-3.218737	0.0013
LEV	8.383419	6.350350	1.320151	0.1869
Effects Specification				
		S.D.	Rho	
Cross-section random		32.07732	0.6757	
Idiosyncratic random		22.22292	0.3243	
Weighted Statistics				
Root MSE	22.29847	R-squared	0.382593	
Mean dependent var	11.50053	Adjusted R-squared	0.381452	
S.D. dependent var	28.48155	S.E. of regression	22.32317	
Sum squared resid	1348963	F-statistic	335.4930	
Durbin-Watson stat	0.892241	Prob(F-statistic)	0.000000	
Unweighted Statistics				
R-squared	0.363152	Mean dependent var	59.09777	
Sum squared resid	4091129	Durbin-Watson stat	0.294197	

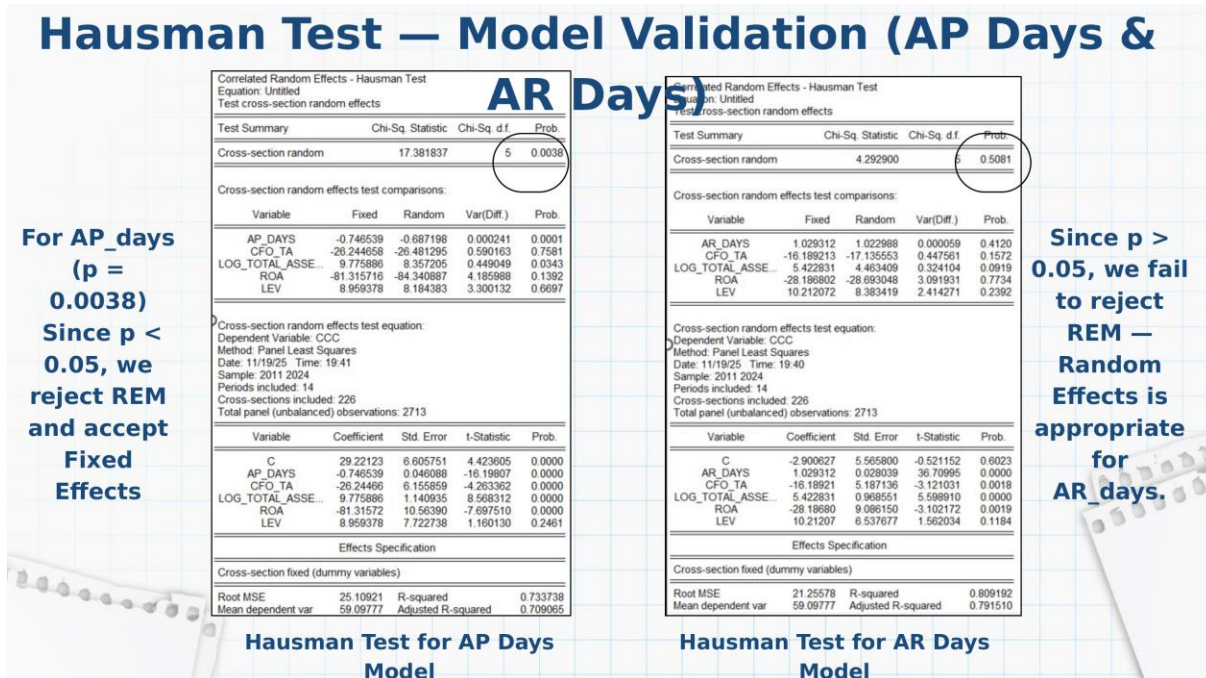
Actual EViews output: Panel EGLS Random Effects estimation for AP_Days and AR_Days models

What this is: A regression that now accounts for firm-specific random effects, re-estimating the impact of AP_Days and AR_Days on CCC with this correction applied.

Layman meaning: The core relationships hold up: AP_Days remains negative (-0.69 , $p < 0.001$) and AR_Days remains positive (1.02 , $p < 0.001$) even after accounting for company-level differences. The AR_Days model explains substantially more of the variation in CCC ($R^2 = 0.38$) than the AP_Days model ($R^2 = 0.14$) at this stage — receivables behavior is the stronger, more explanatory lever.

Actionable for the user: This is the point where the business conclusion becomes reliable enough to act on: AR_Days genuinely deserves more managerial attention than AP_Days, because it explains more of what actually happens to a company's cash cycle.

Output 6 — Hausman Specification Test



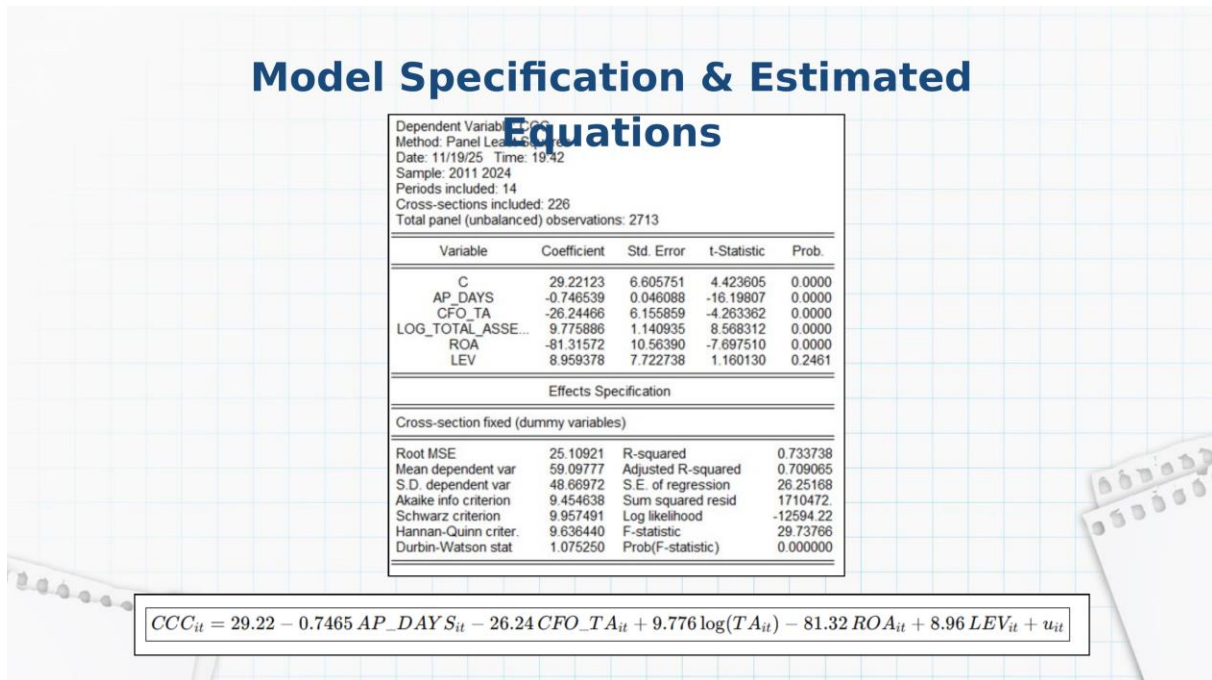
Actual EViews output: Hausman Test comparing Fixed Effects vs Random Effects (both models)

What this is: A test to decide, for each model, whether Random Effects (REM) or Fixed Effects (FE) is the statistically correct specification to report as the final result.

Layman meaning: For the AP_Days model, the test says $p=0.0038$ — significant, meaning REM is not appropriate and Fixed Effects must be used instead (each firm needs its own baseline). For the AR_Days model, the test says $p=0.5081$ — not significant, meaning REM remains valid and appropriate. The two variables literally require different statistical treatments, which is itself a finding: AP_Days behavior is much more firm-specific and idiosyncratic than AR_Days behavior.

Actionable for the user: This explains why a uniform, company-wide payables policy is risky — the statistics show AP_Days effects genuinely differ firm by firm — while a broader, less firm-specific approach to receivables policy is more defensible across the sector.

Output 7 — Final Model Equations



Actual EViews output: Final Fixed Effects (AP_Days) model specification and equation

What this is: The final, correctly-specified equations after all diagnostic testing: AP_Days modelled with Fixed Effects ($R^2=0.734$), AR_Days modelled with Random Effects ($R^2=0.383$).

Layman meaning: Final equations: $CCC = 29.22 - 0.75 \cdot AP_Days - 26.24 \cdot CFO_TA + 9.78 \cdot \log(TA) - 81.32 \cdot ROA + 8.96 \cdot LEV$ (AP model); $CCC = 2.50 + 1.02 \cdot AR_Days - 17.14 \cdot CFO_TA + 4.46 \cdot \log(TA) - 28.69 \cdot ROA + 8.38 \cdot LEV$ (AR model). Both AP_Days and AR_Days remain highly significant ($p < 0.001$) in their final, correctly-specified form — the relationship survived every layer of statistical scrutiny.

Actionable for the user: These are the equations a CFO or working-capital analyst could actually use as a planning tool: for every additional day added to AR_Days, expect CCC to lengthen by roughly one full day; for every additional day added to AP_Days, expect CCC to shorten by about three-quarters of a day. This directly quantifies the trade-off between chasing customer payments faster versus stretching supplier payments further.

Business Takeaway

Receivables management is the single strongest lever for controlling an FMCG firm's cash conversion cycle, with a near 1:1 impact — more so than payables extension. Because AR_Days behaves consistently across firms (Random Effects sufficient) while AP_Days is firm-specific (Fixed Effects required), a sector-wide receivables policy is defensible, but payables strategy should be set individually per company rather than through blanket sector benchmarks.